

Questions 2023-2024 Report writing

Selected questions

Explain the main points of TECHNICAL WRITING INTRODUCTION?

What and Why:

- Technical writing is used to report technical information.
- This is different from creative and other types of writing styles in many ways. We will discuss these later.
- Why is this important? As a scientist/engineer, it is important for you to be able to be able to communicate your work to others in writing.

Voice:

- Technical writing needs to be objective. Why? The purpose of the document is to relay information.
- This is most easily accomplished by reporting "just the fact".
- Adding opinions, personal observations, and other extraneous material can cloud the communication between the author and reader.
- In addition, technical documents are often "serious documents", meaning that much depends on them.
- They need to be concise and specific.

Opinions:

- The author uses his/her words to relay information about facts and events that have happened.
- The author typically does not relay opinions, except in the cases where an unexplained event or phenomena has occurred. In this case, it is

OK for the author to render an opinion (educated, of course) about the situation.

- If this is done, it is important that the author make clear that he or she is offering an opinion and not a fact.
- In a technical paper, the author remains independent from the subject of the article. Technical documents usually do not contain the following words: I, you, we...

Tense:

- As the writer is usually writing about events which have already happened, the tense of the paper is mainly **past tense**.
- However, in some parts of the document, it may be appropriate to use present or future tense.
- For instance, if the author is covering facts that were, are, and forever shall be true, the facts may be referred to in **present tense**.
- If the author is writing about experiments or activities yet to come, **future tense** is appropriate.

What are the Elements of a Scientific Report?

- **Title Page**
- **Table of Contents**
- **Abstract**
- **Introduction**
- **Materials and Methods (Experimental)**
- **Results**
- **Discussion**
- **Conclusion**
- **References**

When Writing a Scientific Report the Introduction is used to provide?

- This provides a summary of the analysis to be undertaken. The purpose of the Introduction is to put the reader in the picture and place the research/experiment within a context.
- The following may be included in the Introduction:
 - Background about the analysis to be carried out.
 - A brief review of previous research (relevant literature) to give a background - paraphrase relevant facts from the scientific literature, citing the sources to support each statement.
 - Reason/s why the research was undertaken.
 - Statement of the hypothesis (an idea or concept that can be tested by experimentation) if there is one.
 - An explanation of the different techniques and why they are used.
 - A statement of the objective/s - what you hope to achieve
- The Introduction is the what and why of the experiment, and should answer the following questions:
 - What was the purpose or objective of the experiment/research?
 - Why was the experiment/research conducted in a particular manner?
 - Why was it important in a broader context?
- The Introduction should not include any results or conclusions.

When Writing a Scientific Report

- **Materials and Methods (Experimental)**

- The Materials and Methods, sometimes called Experimental, is a description of the materials and procedures used - what was done and how.
- Describe the process of preparation of the sample, specifications of the instruments used and techniques employed.
- The Method should include such things as sample size, apparatus or equipment used, experimental conditions, concentrations, times, controls etc.
- While the Method does not need to include minute details (e.g. if you followed a set of written instructions, you may not need to write out the full procedure - state briefly what was done and cite the manual), there needs to be enough detail so that someone could repeat the work.
- Do not keep using the word "then" - the reader will understand that the steps were carried out in the order in which they are written.
- The Method must be written in the **past tense** and the **passive voice**.

Writing a Scientific Report

- State your **interpretation of your findings**, perhaps comparing or contrasting them with the literature. Reflect on your actual data and observations.
- Explain or rationalize errant data or describe possible sources of error and how they may have affected the outcome.
- The Discussion must answer the question **"What do the results mean?"** It is an argument based on the results.

When Writing a Scientific Report the References section is:

- **Citing any references** that you have used, ensuring that each item in the reference list has an in-text citation, and every in-text citation has a full reference in the reference list at the end of your paper.
- Ensure that the references are formatted according to the style required by the journal (or your lecturer/supervisor), and be careful with spelling (the author whose name you misspell may be asked to review the paper!)

What is the purpose and aim of Writing Laboratory Reports?

- **Purpose** aim of your Lab. report is to provide a factual and accurate account of an investigation:
 - what you did
 - what you found
 - what your results mean

Write four important points about scientific writing style

- You should write in complete, grammatically correct sentences.
- **Be concise.** If you can use one word instead of a phrase with two or more words, then choose the one word (get around = avoid).
- **Be objective.** Limit your use of personal pronouns (I, you, we), emotionally loaded words (wonderful, useless, lovely) and casual or ambiguous expressions ('the reaction carried on for 10 minutes').
- **Use technical terms correctly.** Learn what they mean, how to use them and how to spell them. Your lecturer may be able to recommend a good specialist dictionary.
- **Do not use contractions** (isn't, doesn't, it's), While these are common in speech, in formal writing the full form (is not, does not, it is) is expected.

The Simple Lab report

- The simple report is generally only two to five pages long, and usually consists of the following:
- Title page + ID details
- Abstract
- Aims
- Method
- Results
- Discussion
- Conclusion
- References

What is the extended Lab. report? and what are the possible sections of the Extended Lab Report?

- This may be ten pages or longer.
- An extended report consists of the same components of a simple report, plus additional sections, some of which are described on this page.

Possible sections in an extended report

- Table of Contents
- Introduction
- Instrumentation
- Applied Theory
- Aims
- Principles
- Method
- Flow Chart
- Calculations
- Results
- Discussion
- Conclusion
- Appendices

- References

What is an Abstract? Where it is placed in the report or paper? What it include and what it do not include?

- the abstract provides a summary of the entire paper
- it is placed at the beginning of the report
- about 5% of the whole text, including:
 - the problem and its importance
 - what was done (the experiment)
 - how it was done (the method)
 - what resulted (the most important results)
 - what this research contributes to the field (significance)
- Note: The abstract does not include figures or tables.

What the results section provide?

- **The results section:**
- analyses and interprets the results, showing how these relate to the scope of study
- states conclusions about how the results confirm, verify, or support the hypothesis, or refute, negate, or contradict it.

What the conclusion section provides?

- Summarizes the conclusions of the study. Provides an interpretation or summary (not a discussion) of your results. This is normally a brief statement
- It should always reflect the question(s) posed in the Aim(s).
- Sometimes the conclusion is not separate from the discussion, i.e. you may be asked to give a combined "conclusion/discussion".
- Sometimes the conclusion may be required to go after the discussion, in which case it will not be a summary of the results but will be what you conclude based on your discussion.
- This type of conclusion will probably be about a paragraph in length.

What are the main Elements of an abstract in Scientific Report?

- A brief reference to the Materials and Methods.
- A summary of the results and conclusions - a brief but thorough statement of the outcome/s of the experiment.
- If there is a hypothesis, you may state what it is and whether it was supported or refuted.
- The following should not be included in the Abstract:
- Literature citations.
- Formulae and abbreviations, references to tables.
- Although the Abstract comes first in a report, it is best to write it last, after you have the results and conclusions.

What are the main Elements of an Introduction in Scientific Report?

- This provides a summary of the analysis to be undertaken. The purpose of the Introduction is to put the reader in the picture and place the research/experiment within a context.
- The following may be included in the Introduction:
- Background about the analysis to be carried out.
- A brief review of previous research (relevant literature) to give a background - paraphrase relevant facts from the scientific literature, citing the sources to support each statement.
- Reason/s why the research was undertaken.
- Statement of the hypothesis (an idea or concept that can be tested by experimentation) if there is one.
- An explanation of the different techniques and why they are used.
- A statement of the objective/s - what you hope to achieve.
- The Introduction is the what and why of the experiment,
- **The Introduction should answer the following questions:**
 - **What** was the purpose or objective of the experiment/research?
 - **Why** was the experiment/research conducted in a particular manner?
 - **Why** was it important in a broader context?
- The Introduction **should not include** any results or conclusions.

What are The elements of a scientific writing style?

- Punctuation
- Hyphenation
- Italics and Bold
- Fonts and Symbols
- Abbreviations and Acronyms
- Headings, Paragraph Styles, and Lists
- Numbers and Statistics
- Tables
- Figure
- Use of Words

- Grammar
- Flow of Ideas
- Style for Cited Publications
- References

When to use *ITALICS* and **BOLD in scientific writing?**

- Use *italics* for weaker emphasis and bold for strong emphasis. Avoid *italic bold*
- Bold or strong is used for :
 - Important phrases.
 - Highlighting keywords.
- Bold is used to highlight the text and *capture the readers' attention*.
- The bold tag is used for **strong emphasis**. When you feel like emphasizing something, you need to first consider using the italics, only use bold text if you are not satisfied by the emphasis the italics did to your text.
- Put headings in **BOLD UPPER CASE**.
- Put subheadings in **Bold Title Case**.
- Put sub-subheadings in Plain Title Case.
 - Used to highlight quotes (some use block quotes).
 - Used to highlight names of books, blogs, newspapers, ...
 - Conversations or dialogues.
 - Imagination or story (*One day I was walking...*).
 - Foreign words.
- Italics are used in the conversational type of sentences, dialogues, and names.
- They are also used to emphasize a word in a sentence. When it comes to emphasis treat italics as a weaker emphasis when compared to bold.

Use italics first, if it doesn't feel good – use the bold tag

- For example, “It is recommended *not* to drink coffee more than twice a day”.
- Do not use italics for foreign words and abbreviations common in scientific English, such as *ad lib*, *per se*, *et al.*, *via*, *ad hoc*, *post hoc*, *a priori*, *a posteriori*. Put the title of a paper, book, or journal in italics in the body of the text.
- **In the reference list, titles of papers are in normal case**

What are the References?

- Cite any references that you have used, ensuring that each item in the reference list has an in-text citation, and every in-text citation has a full reference in the reference list at the end of your paper.
- Ensure that the references are formatted according to the style required by the journal (or your lecturer/supervisor), and be careful with spelling (the author whose name you misspell may be asked to review the paper!).

What is IEEE Style?

- The [Institute for Electrical and Electronics Engineers \(IEEE\)](#) is a professional organization supporting many branches of engineering, computer science, and information technology. In addition to publishing journals, magazines, and conference proceedings, IEEE also makes many standards for a wide variety of industries.
- IEEE citation style includes in-text citations, numbered in square brackets, which refer to the full citation listed in the reference list at the end of the paper.
- The reference list is organized numerically, not alphabetically. For examples, see the [IEEE Editorial Style Manual](#)

What is In-text Citing in IEEE Style?

- It is not necessary to mention an author's name, pages used, or date of publication in the in-text citation. Instead, refer to the source **with a number in a square bracket**, e.g. [1], that will then correspond to the full citation in your reference list.
- Place bracketed citations within the line of text, before any punctuation, with a space before the first bracket.
- Number your sources as you cite them in the paper.
- Once you have referred to a source and given it a number, continue to use that number as you cite that source throughout the paper.
- The below examples are from [IEEE Style Guide](#).
- Examples of in-text citations:
 - "...end of the line for my research [13]."
 - "This theory was first put forward in 1987 [1]."
 - "Scholtz [2] has argued that..."
 - "Several recent studies [3], [4], [15], [16] have suggested that..."
 - "For example, see [7]."
- When citing multiple sources at once, the preferred method is to list each number separately, in its own brackets, using a comma or dash between numbers, as such: [1], [3], [5] or [1] - [5].

What is a Reference List?

- The Reference List appears at the end of your paper and provides the full citations for all the references you have used.
- List all references numerically in the order they've been cited within the paper, and include the bracketed number at the beginning of each reference.
- Title your list as **References** either centered or aligned left at the top of the page.
- Create a hanging indent for each reference with the bracketed numbers flush with the left side of the page.
- The hanging indent highlights the numerical sequence of your references

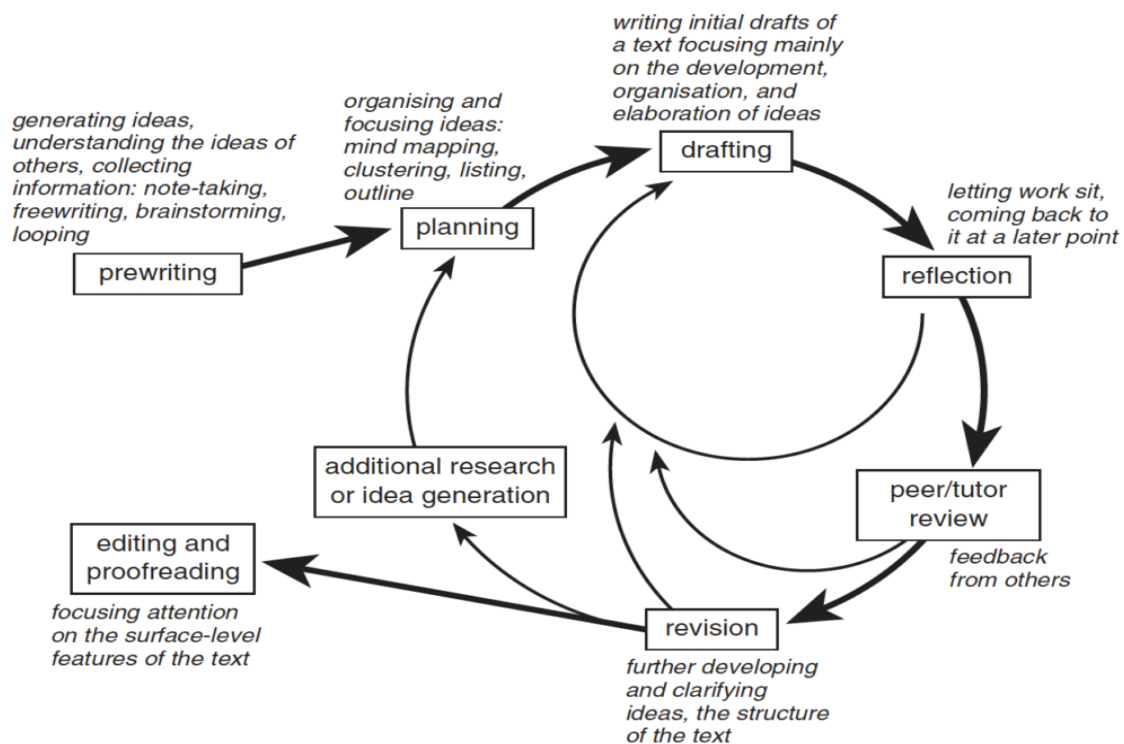
Explain the following elements of each reference in the reference list:

- [4] J. U. Duncombe, "Infrared navigation - Part I: An assessment of feasibility," IEEE Trans. Electron. Devices, vol. ED-11, pp. 34-39, Jan. 1959.
- [7] L. Liu and H. Miao, "A specification based approach to testing polymorphic attributes," in Formal Methods and Software Engineering: Proceedings of the 6th International Conference on Formal Engineering Methods, ICFEM 2004, Seattle, WA, USA, November 8-12, 2004, J. Davies, W. Schulte, M. Barnett, Eds. Berlin: Springer, 2004. pp. 306-19.
- [1] B. Klaus and P. Horn, Robot Vision. Cambridge, MA: MIT Press, 1986.

Write five Article Types, then select one type and explain it.

- 1- Article (original research)
- 2- Review
- 3- Communication
- 4- Commentary
- 5- Short Note
- 6- Case Report
- 7- Proceedings, Extended Abstracts, Meeting Reports
- 8- Interesting Image
- 9- Comment and Reply
- 10- Editorial
- 11- Updates and Amendments: Corrections, Erratums and Addendums
- 12- Expression of Concern
- 13- Retraction

Draw the The technical Writing Process Diagram.



The technical Writing Process Diagram

Write the main Stages of The technical Writing Process

- **Stage 1 – Planning:** spend 40% of your time planning your document (task analysis, thinking, discussing, free-writing, researching, brainstorming, concept mapping, focusing ideas, outlining, etc.)
- **Stage 2 – Drafting:** spend 20% of your time writing a rough draft (quickly getting all your ideas down in print, in more or less complete sentences and paragraphs, in more or less the right order, without agonizing over style or grammar choices)
- **Stage 3 – Revising:** spend 40% of your time revising, editing, and proofreading (polishing your draft, making sure the content is complete)

and well supported, ideas flow logically, formatting meets expectations, expression is grammatically correct and has the appropriate tone and vocabulary).

When to use Nested *ands*?

A nested *and* is really a type of long unpunctuable sentence, though it is often punctuable at the expense of understanding

Describe the good Slide Structure ?

- Use 1-2 slides per minute of your presentation
- Write in point form, not complete sentences
- Include 4-5 points per slide
- Avoid wordiness: use key words and phrases only
- Show one point at a time:
 - Will help audience concentrate on what you are saying
 - Will prevent audience from reading ahead
 - Will help you keep your presentation focused

Describe the Good Colour slide Slide Structure ?

- Use a colour of font that contrasts sharply with the background
 - Ex: blue font on white background
- Use colour to reinforce the logic of your structure
 - Ex: light blue title and dark blue text
- Use colour to emphasize a point
 - But only use this occasionally

Describe the bad Colour slide Slide Structure?

- Using a font colour that does not contrast with the background colour is hard to read
- Using colour for decoration is distracting and annoying.
- Using a different colour for each point is unnecessary
 - Using a different colour for secondary points is also unnecessary
- Trying to be creative can also be bad

Describe the good Background slide Slide Structure?

Background - Good

- Use backgrounds such as this one that are attractive but simple
- Use backgrounds which are light
- Use the same background consistently throughout your presentation

Describe the bad Background slide Slide Structure?

- Avoid backgrounds that are distracting or difficult to read from
- Always be consistent with the background that you use

Graphs - Good

- Use graphs rather than just charts and words
- Data in graphs is easier to comprehend & retain than is raw data
- Trends are easier to visualize in graph form
- Always title your graphs

Graphs - Bad

- Minor gridlines are unnecessary
- Font is too small
- Colours are illogical
- Title is missing
- Shading is distracting

Why you end your presentation with a simple question slide?

- **End your presentation with a simple question slide to:**
 - Invite your audience to ask questions
 - Provide a visual aid during question period
 - Avoid ending a presentation abruptly

Write five BASIC TIPS for effective presentation?

The following points are important for effective presentation:

- **Dress smartly:** don't let your appearance distract from what you are saying.
- **Smile.** Don't hunch up and shuffle your feet. Have an upright posture. Try to appear confident and enthusiastic.
- **Say hello and smile** when you greet the audience: your audience will probably look at you and smile back: an instinctive reaction.
- **Speak clearly**, firmly and confidently as this makes you sound in control.
- **Don't speak** too quickly: you are likely to speed up and raise the pitch of your voice when nervous. Give the audience time to absorb each point.
- **Don't talk in a monotone** the whole time. Lift your head up and address your words to someone near the back of audience.
- **If you think people at the back can't hear**, ask them.
- **Use silence** to emphasise points.
- **Before you make a key point pause:** this tells the audience that something important is coming.
- Nervous speakers tend to gabble on trying to fill every little gap.
- **Keep within the allotted time** for your talk.
- **Eye contact** is crucial to holding the attention of your audience.
 - **Look at everyone in the audience from time to time, not just at your notes or at the PowerPoint slides.**
 - **Try to involve everyone, not just those directly in front of you.**
- **Walk around a little** and gesture تحريك with your hands.
 - **Bad presenters** keep their hands on the podium منصة or in their pockets!
 - **Don't stand in one place** glued to the spot hiding behind the podium!

- Good presenters will walk from side to side and look at different parts of the audience.
- **You could try to involve your audience** by asking them a question.
- **Don't read out your talk**, as this sounds boring and stilted, but refer to brief notes jotted down on small (postcard sized) pieces of card.
- **Don't look at your notes too much** as this suggests insecurity and will prevent you making eye contact with the audience.
- **It's OK to use humour**, in moderation, but better to use anecdotes than to rattle off a string of jokes.
- **Take along a wristwatch to help you keep track of time** – the assessor may cut you off as soon as you have used the time allocated, whether or not you have finished.
- It **can be very helpful to practice at home** in front of a mirror.
- **You can also record your presentation and play it back to yourself: don't judge yourself harshly when you replay this - we always notice our bad points and not the good when hearing or seeing a recording or ourselves! Time how long your talk takes.**
- Run through the talk a few times with a friend.
- It's normal to be a little nervous.
 - This is a good thing as it will make you more energized. Many people have a fear of speaking in public.
 - Practicing will make sure that you are not too anxious. In your mind, visualise yourself giving a confident successful performance.
 - Take a few deep slow breaths before your talk starts and make a conscious effort to speak slowly and clearly.
- **It was found that people who feel embarrassed** are convinced their mistakes are much more noticeable than they really are: we focus on our own behaviour more than other people do and so overestimate it's impact.
- **This is called the spotlight effect.** If you make a mistake, don't apologise too much, just briefly acknowledge the mistake and continue on.
- **Build variety into the talk and break** it up into sections: apparently, the average person has a three minute attention span!

How can you make a good presentation even more effective? write main points only.

- **1. Show your Passion and Connect with your Audience**
 - It's hard to be relaxed and be yourself when you're nervous.
 - But time and again, the great presenters say that the most important thing is to connect with your audience, and the best way to do that is to let your passion for the subject shine through.

- Be honest with the audience about what is important to you and why it matters.
- Be enthusiastic and honest, and the audience will respond.
- **2. Focus on your Audience's Needs**
 - Your presentation needs to be built around what your audience is going to get out of the presentation.
 - As you prepare the presentation, you always need to bear in mind what the audience needs and wants to know, not what you can tell them.
 - While you're giving the presentation, you also need to remain focused on your audience's response, and react to that.
 - **You need to make it easy for your audience to understand and respond.**
- **3. Keep it Simple: Concentrate on your Core Message**
 - When planning your presentation, you should always keep in mind the question:
 - What is the key message (or three key points) for my audience to take away?
 - You should be able to communicate that key message very briefly.
 - Some experts recommend a 30-second 'elevator summary', others that you can write it on the back of a business card, or say it in no more than 15 words.
 - Whichever rule you choose, the important thing is to keep your core message focused and brief.
 - And if what you are planning to say doesn't contribute to that core message, don't say it.
- **4. Smile and Make Eye Contact with your Audience**
 - This sounds very easy, but a surprisingly large number of presenters fail to do it.
 - If you smile and make eye contact, you are [building rapport](#), which helps the audience to connect with you and your subject. It also helps you to feel less nervous, because you are talking to individuals, not to a great mass of unknown people.
 - To help you with this, make sure that you don't turn down all the lights so that only the slide screen is visible. Your audience needs to see you as well as your slides
- **5. Start Strongly**
 - The beginning of your presentation is crucial. You need to grab your audience's attention and hold it.
 - They will give you a few minutes' grace in which to entertain them, before they start to switch off if you're dull. So don't waste that on explaining who you are. Start by entertaining them.
 - Try a story (see tip 7 below), or an attention-grabbing (but useful) image on a slide

- **6. Remember the 10-20-30 Rule for Slideshows**

- This is a tip from Guy Kawasaki of Apple. He suggests that slideshows should:
- Contain no more than 10 slides;
- Last no more than 20 minutes; and
- Use a font size of no less than 30 point.
- This last is particularly important as it stops you trying to put too much information on any one slide. This whole approach avoids the dreaded 'Death by PowerPoint
- As a general rule, slides should be the sideshow to you, the presenter.
- A good set of slides should be no use without the presenter, and they should definitely contain less, rather than more, information, expressed simply.
- If you need to provide more information, create a bespoke handout and give it out after your presentation

- **7. Tell Stories**

- Human beings are programmed to respond to stories.
- Stories help us to pay attention, and also to remember things.
- If you can use stories in your presentation, your audience is more likely to engage and to remember your points afterwards.
- It is a good idea to start with a story, but there is a wider point too: you need your presentation to act like a story.
- Think about what story you are trying to tell your audience, and create your presentation to tell it.

- **8. Use your Voice Effectively**

- The spoken word is actually a pretty inefficient means of communication, because it uses only one of your audience's five senses. That's why presenters tend to use visual aids, too.
- But you can help to make the spoken word better by using your voice effectively.
- Varying the speed at which you talk, and emphasizing changes in pitch and tone all help to make your voice more interesting and hold your audience's attention.
- For more about this, see our page on [Effective Speaking](#)

- **9. Use your Body Too**

- It has been estimated that more than three quarters of communication is non-verbal.
- That means that as well as your tone of voice, your body language is crucial to getting your message across.
- Make sure that you are giving the right messages: body language to avoid includes crossed arms, hands held behind your back or in your pockets, and pacing the stage.

- Make your gestures open and confident, and move naturally around the stage, and among the audience too, if possible.
- **10. Relax, Breathe and Enjoy**
 - If you find presenting difficult, it can be hard to be calm and relaxed about doing it.
 - One option is to start by concentrating on your breathing. Slow it down, and make sure that you're breathing fully. Make sure that you continue to pause for breath occasionally during your presentation too.
 - For more ideas, see our page on [Coping with Presentation Nerves](#).
 - If you can bring yourself to relax, you will almost certainly present better. If you can actually start to enjoy yourself, your audience will respond to that, and engage better. Your presentations will improve exponentially, and so will your confidence. It's well worth a try.

What is plagiarism?

- Specifically, plagiarism is:
 - copying phrases and passages word-for-word without quotation marks and without a reference to the author; this includes but is not limited to books, journals, reports, theses, websites, conference papers and course notes
 - paraphrasing an author's work and presenting it without a reference; this includes but is not limited to sentences, paragraphs, ideas and themes
 - copying any part of another students' work
 - submitting items of assessment that are written in conjunction with other students (without prior permission of the relevant staff member)

How to avoid plagiarism?

- submitting a piece of work has already been submitted for assessment in another course
- presenting other people's designs and images as your own work
- submitting work as your own that someone else has done for you.

- **If you have used someone else's work without acknowledging your source, you have plagiarised.**
- **Plagiarism can sometimes be the result of poor note taking, or paraphrasing without properly citing the reference.**
- **You can avoid plagiarism by:**
 - citing your references
 - referencing correctly
 - recording direct quotes and paraphrases correctly when note taking.
- **Quotes**
 - When you use the exact words, ideas or images of another person, you are quoting the author.
 - If you do not use quotation marks around the original author's direct words and cite the [reference](#), you are plagiarising.

Paraphrasing

- Poor note taking can lead to plagiarism. You should always take care to:
- record all reference information correctly
- use quotation marks exactly as in the original paraphrase correctly
- clearly distinguish your own ideas from the ideas of other authors and researchers.

Answer the following

1- Revise the space between the following numbers and unit, write which is acceptable style and which is not. correct the wrong ones.

75kg. 75 %. .32.

Mark the correct answer (✓),

- a. () Illustrations in technical writing may include creating Illustrations such as: Tables, Bar and Column Graphs, Circle Graphs, Line Graphs , Flow Charts, and Diagrams
- b. () Parts of Technical Report are : Introduction, Summary, Conclusion, Story, Recommendation and Personnel.
- c. () You don't start a sentence with numbers greater than ninety-nine. one, two, three... nine, 10, 11, 12... Exceptions: a 2-m tape measure; 3 million.
- d. () An impersonal style uses the passive voice, the third person rather than the first person (it rather than I or we) and things rather than people as subjects of sentences.
- e. () While writing a technical report, it is not necessary to make acknowledgment.
- f. () Correct referencing is important for a number of reasons including to support your arguments
- g. () Citations can be of two basic types: information-prominent and topic-prominent.
- h. () Abstract Determines whether a particular Style is to be expected
- i. () Sentence Length is not important
- j. () A paragraph contains only one topic sentence.

•

• Question 4

A. Write some basic PowerPoint Presentation Style Tips?

B. How can you make a good presentation even more effective?

•

Answer The following questions:

1. What are the outlines or the strategies to help you with various writing tasks?
2. Write some basic tips for making job technical presentation?
3. How can you make a good presentation even more effective?
4. Explain why literature reviews are important to scientists?
5. Write steps on how to write the literature review?

6. What is plagiarism?
7. How to avoid plagiarism

• **Question 1**

- A. What are the common components of a Simple TECHNICAL REPORT?
- B. What is a table of contents in technical writing? where it is usually used and where it is not?

• **Question 2**

- A. Explain the contents of the following citations.
 - a. [5] Hadhoud M. M. and Thomas D. W., "The two dimensional adaptive LMS algorithm", IEEE Trans. Circuits Systems, Vol. 35, No. 5, pp. 485- 494, 1988.
 - b. [20] Hayken S., Adaptive Filter Theory, NJ, Prentice Hall, 4th Edition, 2002.
 - c. [17] Gollamudi S., Nagaraj S., Kapoor S. and Huang Y. F., "Set-Membership Filtering and a set membership normalized LMS Algorithm with an Adaptive Step-Size", IEEE signal processing Lett, Vol. 5, No. 5, pp. 111-114, May 1998.
 - d. [10] Muneyasu M., Uemoto E. and Hinamoto T., "A novel 2-d adaptive filter based on the 1-d RLS algorithm", IEEE International Symposium on Circuits and Systems, Hong Kong, pp. 2317- 2320, June 1997
- B. Write what are the purposes of Lab report writing?

B. • **Question 3**

- In Basic Principles of **Effective writing Style** mark (✓) to the correct item:
 - k. () Illustrations in technical writing may include creating Illustrations such as: Tables, Bar and Column Graphs, Circle Graphs, Line Graphs , Flow Charts, and Diagrams
 - l. () Parts of Technical Report are : Introduction, Summary, Conclusion, Story, Recommendation and Personnel.
 - m. () You don't start a sentence with numbers greater than ninety-nine. one, two, three... nine, 10, 11, 12... Exceptions: a 2-m tape measure; 3 million.
 - n. () An impersonal style uses the passive voice, the third person rather than the first person (it rather than I or we) and things rather than people as subjects of sentences.
 - o. () While writing a technical report, it is not necessary to make acknowledgment.
 - p. () Correct referencing is important for a number of reasons including to support your arguments
 - q. () Citations can be of two basic types: information-prominent and topic-prominent.
 - r. () Determine Whether a Particular Style Will Be Expected

- s. () Sentence Length is not important
- t. () Write to Express, Not to Impress

C. How can you make a good presentation even more effective?

D. What is The Extended Lab Report, what are the sections of the extended lab report?

- The Extended Lab Report may be ten pages or longer. An extended report consists of the same components of a simple report, plus additional sections, some of which are described on the following points:.

-
- Possible sections in an extended report
- Table of Contents
- Introduction
- Instrumentation
- Applied Theory
- Aims
- Principles
- Reagents
- Method
- Flow Chart
- Calculations
- Results
- Conclusion
- Discussion
- Appendices
- References

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• **Question 5**

- Write steps on how to write the literature review?

A. What is plagiarism? How to avoid plagiarism?

•

C. What are the elements of a **Simple Scientific Report**?

D. What is a table of contents in technical writing? where it is usually used and where it is not?

•

B. Write the purposes of the Introduction?, what are the main points to be included in the introduction?

• **Question 3**

- In Basic Principles of **Effective writing Style** mark (✓) to the correct item:

•

• **Question 5**

B. Why Literature reviews are important to scientists? what Literature reviews will give to the reader?

C. What is plagiarism? How to avoid plagiarism?

•

•

- **Question 1**

E. What are the elements of a Simple **Scientific Report**?

- Elements of a Scientific Report
- Title Page
- Table of Contents
- Abstract
- Introduction
- Materials and Methods (Experimental)
- Results
- Discussion
- Conclusion
- References
-

F. What is a table of contents in technical writing? where it is usually used and where it is not?

B. Write the purposes of the Introduction?, what are the main points to be included in the introduction?

- is to put the reader in the picture and place the research/experiment within a context. The following may be included in the Introduction:
- Background about the analysis to be carried out.
- A brief review of previous research (relevant literature) to give a background - paraphrase relevant facts from the scientific literature, citing the sources to support each statement.
- Reason/s why the research was undertaken.
- Statement of the hypothesis (an idea or concept that can be tested by experimentation) if there is one.
- An explanation of the different techniques and why they are used.
- A statement of the objective/s - what you hope to achieve.
- The Introduction is the what and why of the experiment, and should answer the following questions:
 - a. What was the purpose or objective of the experiment/research?
 - b. Why was the experiment/research conducted in a particular manner?
 - c. Why was it important in a broader context?
- The Introduction should not include any results or conclusions.
-
-
-

- **Question 4**

- **A- What makes a good Science communicator?**
-

- Scientists write, among other things, to inform the public, to persuade government and industry to fund research, and to communicate results, innovations and discoveries to fellow academics, industry, and public

audiences. Whatever the form of communication used – oral presentation, report, academic paper, website or news item – the good Science writer:

- thinks objectively and thoroughly
- researches carefully
- keeps good records and notes
- writes clearly, concisely and accurately
- considers the background of the audience
- uses the appropriate format for the type of writing involved
- presents the material neatly
- takes care to acknowledge all sources of information.

B- Answer the following:

- 1- Complete the sentence using the modal verbs in a sentence: **need to - appears - indicates**
- The data to support the hypothesis, but further sampling would be undertaken to increase the reliability of the final result.
- 2- Complete the sentence using the softening words: seems to - appears to - suggest
- The results shown in Figure 1 that stratified sampling by gender...
- The evidence indicate that the flow rate of NaOH does not significantly affect...
- 3- Use the following words to complete the sentence: overrun - catch - increase
- When people too many fish, the coral pests and eventually the coral colonies.
- The data **appears** to support the hypothesis, but further sampling would **need to** be undertaken to increase the reliability of the final result.
- 2- Complete the sentence using the softening words: seems to - appears to - suggest
- The results shown in Figure 1 **suggest** that stratified sampling by gender...
- The evidence **seems to** indicate that the flow rate of NaOH does not significantly affect...
- 3- Use the following words to complete the sentence: overrun - catch - increase
- When people **catch** too many fish, the coral pests **increase** and eventually **overrun** the coral colonies.

•

D. How can you make a good presentation even more effective?

•

- Ideas to help you to make a good presentation even more effective.
- 1. Show your Passion and Connect with your Audience
- 2. Focus on your Audience's Needs
- 3. Keep it Simple: Concentrate on your Core Message
- 4. Smile and Make Eye Contact with your Audience
- 5. Start Strongly

6. Remember the 10-20-30 Rule for Slideshows
7. Tell Stories
8. Human beings are programmed to respond to stories.
9. Use your Voice Effectively
10. Use your Body Too
11. Relax, Breathe and Enjoy

-
-
-

• **Question 5**

D. Why Literature reviews are important to scientists? what Literature reviews will give to the reader?

- Literature reviews are important to scientists because
- scientific knowledge is not static
- scientific knowledge changes as scientists conduct their research, replicate the research of other scientists, and report their findings
- it is important to understand how knowledge in a particular discipline is changing
- it is very important to be aware of the controversies in the discipline.
- A literature review will give the reader
 - a. a clear statement of the topic or problem
 - b. a clear presentation of the range of research on the topic
 - c. an evaluation of the research
 - d. an indication of what further research is necessary.
- It is not enough to summarise the research literature. You need to show you understand the relevance and significance of the different publications, and how they relate to each other.

-

E. What is plagiarism? How to avoid plagiarism?

- Plagiarism is the presentation of another person's work, ideas, or creations as your own, without acknowledging where those ideas came from. In other words - you take credit for someone else's work. In academic writing, this is the same as cheating on an exam. Specifically, plagiarism is:
 - a. copying phrases and passages word-for-word without quotation marks and without a reference to the author; this includes but is not limited to books, journals, reports, theses, websites, conference papers and course notes
 - b. paraphrasing an author's work and presenting it without a reference; this includes but is not limited to sentences, paragraphs, ideas and themes
 - c. copying any part of another students' work
 - d. submitting items of assessment that are written in conjunction with other students (without prior permission of the relevant staff member)
 - e. submitting a piece of work has already been submitted for assessment in another course
 - f. presenting other people's designs and images as your own work

- g. submitting work as your own that someone else has done for you.
-
- Plagiarism can sometimes be the result of poor note taking, or paraphrasing without properly citing the reference. You can avoid plagiarism by:
 - a. citing your references
 - b. referencing correctly
 - c. recording direct quotes and paraphrases correctly when note taking.
-
-
-

Mid Term retake

Answer the following questions:

1- The following should not be included in the Abstract:

- a. Objectives (as outlined in the Introduction) and scope of the investigation.
- b. A brief reference to the Materials and Methods.
- c. A summary of the results and conclusions - a brief but thorough statement of the outcome/s of the experiment.
- d. If there is a hypothesis, you may state what it is and whether it was supported or refuted.
- e. Literature citations.
- f. Formulae and abbreviations, references to tables.

2- Which of the following statements is NOT true?

- a. As long as you cite your sources, you have permission to use someone else's material.
- b. Your reputation for integrity is considered an important credential for your job.
- c. Your company could be the subject of a criminal case or civil suit if you plagiarize.
- d. In cases of copyright violation, your intentions are immaterial.
- e. Permission is not needed for material in the public domain

3- Explain when **ABBREVIATIONS AND ACRONYMS** are justified?

Selected questions

1- What are the main Elements of a Scientific Report?

2- Write what should be included in the abstract and what should not?

3- Define technical writing.

5- What are the responsibilities of professional/technical writers that cause ethical conflicts?

6- Define and discuss “style of writing ” and its role and importance in technical/professional writing, both professionally and pedagogically. Include at least a brief discussion about the differences between style and grammar.

7- Write what are the purposes of Lab report writing.

8- Write a few important points you should consider about scientific writing style

9- Explain the contents of the Conclusion of a Lab report.

10- Explain why the table of contents is necessary in some reports?

11- What are the characteristics (standards) of technical writing?
when to use ITALICS AND BOLD in technical style writing?

12- Explain when ABBREVIATIONS AND ACRONYMS are justified?

13- Explain the rules of writing FIGURES in technical writing style?

14- What makes a good Science communicator?

- Answer all the following questions
-
- Question 1 20 marks
-
- A- What is a scientific report document?
-
- A scientific report is a document that describes the process, progress, and or results of technical or scientific research or the state of a technical or scientific research problem. It might also include recommendations and conclusion of the research
-
- B- What are the elements of a Scientific Report?
-
- Elements of a Scientific Report
- Title Page

- Table of Contents
- Abstract
- Introduction
- Materials and Methods (Experimental)
- Results
- Discussion
- Conclusion
- References
-
- C- Title Page
- The title page is used to convey the following:
 - • Title
 - • Authors
 - • Author affiliation
 - • Date published
 - • Other... Depending on required format, more information may be placed on this page
-
- Abstract
- The abstract is a concise report on the important contents of the document. Abstracts are usually one or two paragraphs. The purpose of the abstract is to provide a quick review of the important contents of the document so that others can determine if they should invest time reading the rest of the document.
-
- Table of Contents
- (Usually for books and thesis, articles typically do not have this). The table of contents is a listing of the main headings of the document. This includes the abstract, nomenclature, chapters, chapter sections, references, and appendices.
-
- Nomenclature
- The nomenclature is a tabulated listing of the variables (and their units) that will be used in the document. Chapters/Sections The chapters/sections make up the body of the document. They include the detailed information the author wants to convey.
-
- References
- This section lists the books, magazines, conversations, websites, and other sources of information you used in writing your document.
-
- Appendices
- Appendices contain information which is important, though in most cases secondary to the purpose of the document. For instance tabulations of scientific data, detailed calculations, and similar material are often good candidates for appendices.
-
- Question 2
- A- Writ what are the following and how they are used in Sceintefic writing stile?

- 1- Punctuation
- Insert a comma wherever there would be a slight pause between words or phrases in the spoken sentence.
- Insert a semicolon between two parts of a sentence; the proviso is that both parts must be able to stand alone as separate sentences.
- Use a colon to introduce an explanation or an example of something: here is an example. If there are several simple explanations or examples, separate them with commas; otherwise, use semicolons.
- Avoid excessive use of parentheses (). Use them to make an aside (an extra remark) only if commas could be confusing. Never use parentheses within parentheses: find another way of saying it.
- Use brackets [] for material inserted into a quotation and ellipsis (three dots) for material omitted: According to Smith (1999), "few such [descriptive] studies were done... before 1950."
- Use dashes--two hyphens with no spaces anywhere--for emphatic asides.
- Use one or two spaces after a period, colon, or semicolon. Note, though, that Web browsers delete more than one space unless you make them non-breaking spaces.
- Use double quotation marks (") for speech and verbatim quotations.
-
- 2- Tables
- Tables are used as a means to quickly compare information side-by-side.
- Tables have a heading which precedes the table, usually in a font that is bold and larger than the text of the table.
- The headings contain information about the units for the data.
-
- 3- Style for Cited Publications
- Cite any references that you have used, ensuring that each item in the reference list has an in-text citation, and every in-text citation has a full reference in the reference list at the end of your paper.
- Ensure that the references are formatted according to the style required by the journal (or your lecturer/supervisor), and be careful with spelling (the author whose name you misspell may be asked to review the paper!)
-
-
- B- How Article Types are assigned their names?
- All articles are assigned a type, dependent on the content of the article.
- It is useful to readers, to inform them of what style of content to expect (original research, review, commentary, etc.) and, for indexing services when applying filters to search results.
-
-
- Question 3
- A- Write 4 Article Types?
- Article

The most popular type of published paper, articles report original research results.

They report a complete piece of research, typically including introduction (background and motivation), methods, experimental results, and discussion.

- Review papers

consist of concise and precise updates on the latest progress made in a given area of research.

They should be comprehensive and objective and not, for example, only report research results from a single research group or country.

- Commentary

Briefer than a review paper, a Commentary is usually an invited paper that gives an opinion on a specific topic.

They are not as comprehensive as a review paper but should highlight the most significant papers and milestones in the subject area.

-

- Case Report

case study is: An analysis of a project, campaign, or company that: identifies a situation, recommended solutions, implementation actions, and

identification of those factors that contributed to failure or success

Common in medical journals, case reports present detailed information on the symptoms, signs, diagnosis, treatment (including all types of interventions), and outcomes of an individual patient.

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- B- Explain what is the Writing Processes?

Just as we use design processes to creatively solve complex problems, we use writing processes to create complex documents.

In both cases, there are steps or stages, but we don't always proceed directly from one step to next in a chronological manner.

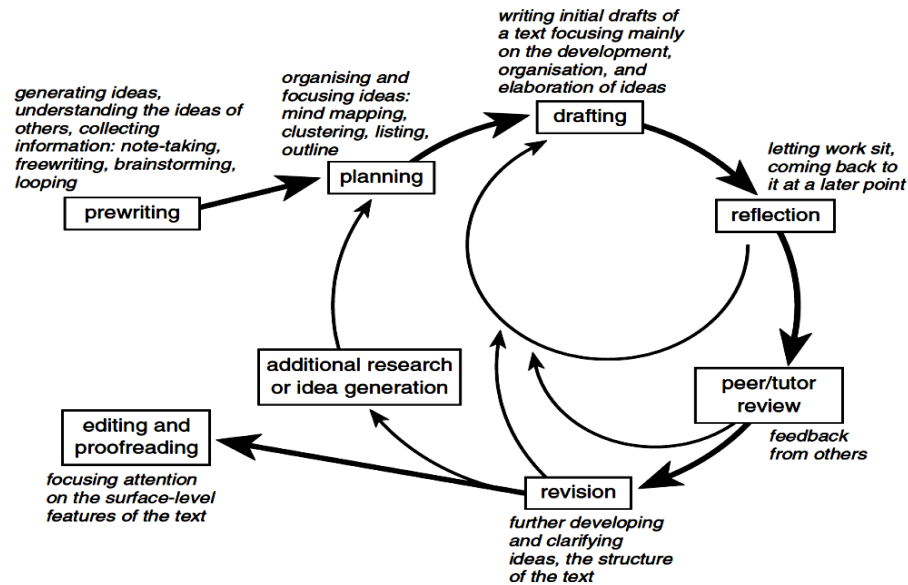
These processes are often iterative, meaning we might return to previous stages in the process from time to time.

The more complex the task, the more iteration might be needed.

Examine the Writing Process (Figure 2) diagrams below.

-

- C- Draw the writing process diagram



-
-
- Question 4
- A- What is the explanation of the 40-20-40 Writing Process?
- The 40:20:40 rule of writing:
40% of the effort going into a piece of writing is preparation
20% of the effort goes to the first draft
40% of the effort goes to revision, rewriting, editing, and proof-reading
-
- B- Writ examples for in-text citations using the IEEE style: differen
forms for 1 reference, the form for 2 references, and the form for more
than two references
- Examples of in-text citations:
- "...end of the line for my research [13]."
- "This theory was first put forward in 1987 [1]."
- "Scholtz [2] has argued that..."
- "Several recent studies [3], [4], [15], [16] have suggested that...."
- "For example, see [7]."
-
- C- What makes a good Science writer?
- the good Science writer:
thinks objectively and thoroughly
researches carefully
keeps good records and notes
writes clearly, concisely and accurately
considers the background of the audience
uses the appropriate format for the type of writing involved
presents the material neatly
takes care to acknowledge all sources of information.
-
- Question 5
-

- A- In scientific writing, Writ examples of the following:
- Limiting words: possible or possibly, probable or probably, likely.
- modal verbs: may or may be, might, might be or might have been,
- could be, could have been, Would, would have been .
-
- B- Citations can be of two basic types, what are these typs
- information-prominent
- author-prominent
-
- C- Tips to be Covered while making presentation?
- Outlines
- Slide Structure
- Fonts
- Colour
- Background
- Graphs
- Spelling and Grammar
- Conclusions
- Questions
-